

AI USE CASE INTEGRATION IN TEACHING-LEARNING

Part A: AI Use Case Profile

1. **Use Case ID:** KJS-AGR-01
2. **Use Case Title:** Irrigation Advisory System for Sugarcane Crop using AI and Sensor-based Technology
3. **AI Application Vertical:** Agriculture
4. **Collaborating Organization:** K J Somaiya Applied Agricultural Research (KIAAR), Sameerwadi, Karnataka, Godavari Biorefineries Ltd. (GBL)
5. **Faculty Owner(s):**
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 - Dr. Radhika Kotecha, Professor, Department of Information Technology, K J Somaiya Institute of Technology
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 - Dr. Balwant Singh, Assistant Professor, Department of Electronics Engineering, K J Somaiya School of Engineering
6. **Beneficiaries:** Farmers (approx. 18,000–25,000 sugarcane farmers associated with GBL and KIAAR, Cane department supervisors and field assistants, Agricultural researchers and agronomists, Sugar factory management for planning cane availability and productivity.
7. **Problem Statement:** Sugarcane cultivation requires large quantities of water, and irrigation scheduling is generally based on farmers' experience rather than scientific recommendations. Due to inadequate knowledge of crop water requirements, soil moisture conditions, weather forecasts, and irrigation timing, farmers often practice over-irrigation or delayed irrigation, resulting in:
 - Water wastage
 - Reduced crop productivity
 - Yield loss due to moisture stress
 - Increased energy consumption
 - Higher cultivation costs
 - Nutrient losses because of improper fertigation
 - Inefficient use of electricity supplied for irrigation pumps
 - Difficulty in monitoring thousands of spatially distributed farms

There is a need for an AI-driven irrigation advisory system that integrates field sensors, weather forecasts, geospatial information, and machine learning models to provide plot-specific irrigation recommendations and decision support to farmers in a simple and understandable manner.

8. Background and Context:

The KIAAR and GBL ecosystem supports nearly 18,000–25,000 sugarcane farmers spread across Northern Karnataka. Through previous geofencing initiatives, thousands of sugarcane plots have already been digitized and spatially mapped. Soil testing records, farmer details, and crop information are available through existing digital platforms. However, irrigation decisions are

still largely manual and depend on experience rather than data-driven approaches. Variations in soil types, crop stages, weather conditions, rainfall events, and water availability necessitate dynamic irrigation recommendations.

Recent advances in IoT and sensor technologies, Remote sensing, Artificial Intelligence and Machine Learning, Weather forecasting systems, Geospatial analytics, Mobile applications and dashboards, etc. provide an opportunity to build a smart irrigation advisory system capable of delivering personalized recommendations at the plot level.

The proposed system aims to combine sensor observations, satellite-derived information, weather forecasts, and AI models to improve irrigation efficiency, enhance water productivity, and promote sustainable sugarcane cultivation.

9. Objectives:

- 1) Develop an AI-enabled irrigation advisory system for sugarcane cultivation that integrates field observations, weather information, and crop-specific requirements to support informed irrigation decision-making.
- 2) Monitor soil moisture, environmental conditions, and crop status using sensor networks and geospatial technologies to enable plot-level assessment and precision irrigation management.
- 3) Generate real-time, forecast-based, and plot-specific irrigation and fertigation recommendations to optimize irrigation scheduling, improve resource utilization, and support operational constraints such as electricity availability.
- 4) Develop farmer-friendly web and mobile platforms for advisory dissemination, visualization of geofenced plots, crop information management, and decision support.
- 5) Enhance water use efficiency, reduce yield losses due to water stress, and improve the sustainability and climate resilience of sugarcane farming practices.

10. Solution Approach:

The proposed system will integrate multiple data sources and AI models to provide intelligent irrigation recommendations: IoT Sensors, Geospatial Data, Weather Data, and Farm Information and develop the below listed AI Models for Prediction:

- Next Irrigation Date Prediction
- Irrigation Duration Prediction
- Crop Water Requirement Prediction
- Water Stress Probability Model
- Rainfall-Adjusted Irrigation Recommendation
- Yield Loss Prediction Due to Delayed Irrigation
- Pump Scheduling Optimization
- Fertigation Recommendation Model
- Disease and Water Stress Forecasting
- Yield Prediction Model
- Farmer-Friendly Advisory Generation Model (LLM-Based)

11. Operational Challenges:

A. Sensor-related

- Sensor calibration and maintenance.
- Power supply and battery life.
- Data communication failures.
- Missing or noisy sensor data.

B. AI Challenges

- Limited labeled datasets.
- Model generalization across soil types.
- Seasonal variability.

C. Geospatial Challenges

- Accurate plot mapping.
- Dynamic crop boundaries.
- Multi-source data integration.

D. Weather Challenges

- Reliability of rainfall forecasts.
- Micro-climatic variations.

E. Implementation Challenges

- Connectivity issues in villages.
- Farmer adoption and trust.
- Device maintenance.
- Training requirements.
- Scalability to 25,000 farms.
- Continuous model updating.
- Integration with existing STEPS (Somaiya Technology for Efficient Plot Survey) platforms.

12. Dataset: <https://drive.google.com/file/d/1oMNs66R7JiOSif-vCCmwaLvoj5c7bqFm/view?usp=sharing>

13. High-Level Workflow:

- 1) Data Acquisition: Collection of field, crop, soil, weather, and geospatial information through mobile applications, databases, APIs, sensor networks, IoT devices, satellite imagery, and GIS-based plot digitization.
- 2) Data Integration: Integration of sensor observations, weather data, satellite imagery, soil reports, farmer records, and historical yield information into a unified agricultural data platform.
- 3) AI Analytics: Development of AI/ML models for soil moisture forecasting, irrigation scheduling, water stress prediction, fertigation planning, pump scheduling optimization, yield loss estimation, and yield prediction.
- 4) Decision Support: Generation of plot-specific irrigation and fertigation recommendations based on field conditions, crop requirements, weather information, and predictive analytics.
- 5) Dashboard and Advisory Platform: Delivery of recommendations through web and mobile applications featuring farmer dashboards, spatial visualization of plots, irrigation and weather information, soil and crop health monitoring, fertigation guidance, yield analytics, and alert generation.

Details of workflow activities:

Workflow Stage	Activities	Details
1. Field Data Acquisition	Sensor deployment, IoT devices, data collection	Design sensor interfaces, communication modules, and wireless data acquisition systems (EXTC)
		Development of embedded systems and edge devices for sensor integration (ECE)
		Communication protocols and IoT gateway connectivity (CCE)
		Smart sensing systems and autonomous monitoring solutions (RAI)

2. Data Transmission and Integration	Data ingestion, cloud connectivity, database creation	Development of APIs, cloud infrastructure, database management and integration platforms (IT)
		Backend systems, middleware, and data synchronization mechanisms (CE)
		Enterprise applications and process workflows for agricultural data management (CSBS)
3. Data Pre-processing and Feature Engineering	Data cleaning, normalization, feature extraction	Data pre-processing, feature engineering, and preparation of datasets for machine learning models (IT, CE, AI&DS)
		Development of data pipelines and automated processing frameworks (IT, CE)
4. AI Analytics and Predictive Modeling	Prediction of irrigation date, duration, water stress, yield loss, pump scheduling	Development and validation of machine learning and deep learning models (IT, CE, AI&DS)
		Intelligent decision-making algorithms and optimization models (RAI)
		AI model deployment and inference frameworks (CE)
5. Decision Support Engine	Rule-based recommendations and optimization	Design of business logic and recommendation engines (IT)
		Workflow optimization and integration with enterprise applications (CSBS)
		AI-assisted decision-making and predictive analytics (IT, CE, AI&DS)
6. Generative AI Advisory Layer	Conversion of predictions into farmer-friendly advisories	Large Language Models (LLMs), multilingual advisory generation, chatbot development (IT, CE, AI&DS)
		Development of AI service integration and API interfaces (IT, CE)
		Integration of AI models into applications and scalable deployment (CE)
7. Dashboard and Mobile Application Development	Web and mobile interfaces, visualization, GIS maps	Full-stack web and mobile application development (IT, CE)
		Backend services, APIs, authentication, and database integration (CE)
		User workflows, reporting systems, and management dashboards (CSBS)
		AI visualization and analytics dashboards (IT, CE, AI&DS)
8. Smart Irrigation Automation	Automated pump operation and scheduling	Embedded controllers and smart irrigation hardware (ECE)
		Wireless communication and control systems (EXTC)
		Intelligent autonomous irrigation control systems (RAI)
		IoT communication protocols and remote monitoring (CCE)
9. Farmer Interaction and Support Systems	Notifications, chatbot, advisory dissemination	Mobile applications and notification services (IT, CE)
		Conversational AI and personalized recommendations (IT, CE, AI&DS)
		User experience and service workflows (CSBS)
10. Continuous Learning and	Feedback collection, model	Continuous learning frameworks and model enhancement (IT, CE)

System Improvement	retraining, performance monitoring	Monitoring platforms and cloud services (IT, CE)
		System scalability and performance optimization (IT, CE)
		Adaptive and self-learning intelligent systems (RAI)

14. Expected Outputs:

- IoT System for data acquisition
- AI-enabled irrigation advisory platform.
- Farmer dashboard.
- Multi-source integrated agricultural data repository.
- Real-time irrigation recommendation system.
- Forecasting and alert mechanisms.
- LLM-based multilingual advisory module.

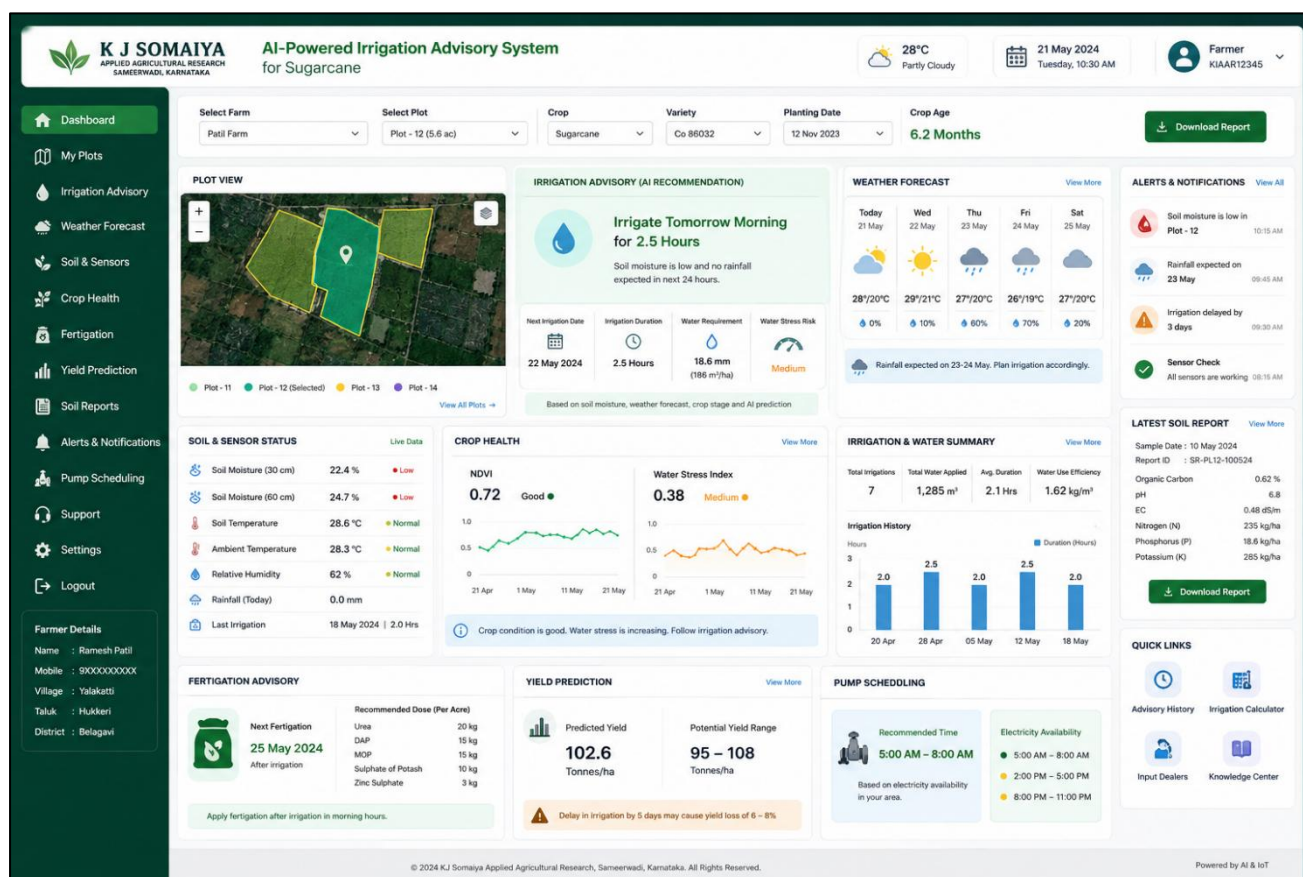


Fig. Software Output Sample

15. Modern AI Governance and Paradigms:

Area	Relevance	Recommended Consideration
Privacy & Data Protection	High	Ensure secure handling of farmer information, plot boundaries, sensor data, and soil reports through role-based access control, encryption, secure APIs, and data anonymization wherever required. Compliance with Digital Personal Data Protection (DPDP) Act and institutional data policies should be ensured.
AI Security & Trustworthiness	High	Protect AI models and IoT infrastructure against cyber threats, unauthorized access, data tampering, and adversarial attacks. Implement secure communication protocols, model validation

		mechanisms, audit logs, and periodic security assessments to maintain reliability and trust in advisory recommendations.
Responsible AI	High	Ensure transparency, fairness, explainability, and accountability in AI-generated irrigation advisories. Models should provide interpretable recommendations and avoid bias arising from variations in soil types, crop varieties, or geographic regions. Continuous monitoring and validation should be carried out before large-scale deployment.
Generative AI / LLMs	High	Utilize Large Language Models (LLMs) to convert technical predictions into farmer-friendly multilingual advisories in English, Kannada, Marathi, and Hindi. Examples include recommendations such as “Irrigate tomorrow morning for 2.5 hours” or “Rainfall is expected; irrigation is not required today.” LLMs can also support chatbot-based query handling and knowledge dissemination.
Agentic AI	Medium to High	Intelligent AI agents may autonomously analyze sensor data, weather forecasts, crop stage, and electricity availability to generate personalized irrigation schedules, fertigation recommendations, and alert notifications. Agentic workflows can support proactive decision-making and continuous monitoring across thousands of farms.
Human-in-the-Loop Decision Making	High	AI recommendations should complement, not replace, agronomists, field supervisors, and farmers. Critical decisions regarding irrigation, fertigation, and stress management should incorporate expert validation and farmer feedback. Mechanisms for manual override and advisory acceptance should be provided within the dashboard and mobile application.
Regulatory / Governance Considerations	Medium to High	Ensure adherence to national policies and standards related to AI, geospatial data, IoT deployment, and digital agriculture. Establish governance mechanisms for data ownership, consent management, data sharing, model versioning, and periodic audits. Clearly define responsibilities among KIAAR, GBL, technology providers, and farmers to ensure ethical and sustainable use of AI technologies.
